CENTRAL COALFIELD LIMITED

PRESENTED BY RAJ LAKRA

COLLEGE ID – BETCH/10783/22

SEMESTER -V

DURATION- 4 WEEKS

STUDENT OF

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CERTIFICATE

This is to Certify that Raj Lakra a student of BIT MESRA ,

RANCHI with Roll No: BTECH/10783/22 has successfully completed the training of 4 weeks at Central Coalfield Limited.

Signature of assistant manager

(civil)

Aman Ujjwal

(Civil Department)

Signature of General Manager

(civil)/H0D

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(Civil Department)

ACKNOWLEDGEMENT

I express my deep gratitude to Mr. Alok Chandra Moharana , General Manager (Civil), CENTRAL COALFIELDS LTD, Darbhanga House Ranchi for his valuable suggestions and guidance rendered in giving shape and coherence to this endeavour. I am also thankful to Aman Sir, for his kind support and guidance to successfully complete my training. I acknowledge the cooperation received from all the employees of Central Coalfields Limited, Ranchi as a whole for providing me the opportunity to learn from their systematic approach of accomplishing the work.

Regards,

RAJ LAKRA
CIVIL ENGINEERING

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CCL

CENTRAL COALFIELDS LIMITED

Central Coalfields Limited (CCL) is a subsidiary of Coal India Limited (CIL), an undertaking of the Government of India. CCL was established in 1956 as National Coal Development Corporation Limited and is a Category-I Mini Ratna company since October 2007. CCL manages the nationalized coal mines of the Coal Mines Authority, Central division. CCL is headquartered at Darbhanga House, Ranchi, Jharkhand.



Infrastructure

CCL owns and operates the following:

I. **43 operative mines (5 underground & 38 opencast)**
II. **7 operating coalfields. These coalfields are located in East** Bokaro, West Bokaro, North Karanpura, South Karanpura, Ramgarh, Giridih and Hutar.
III. 7 washeries
    a. 4 operational coking coal washeries at Rajrappa, Kedla, Kathara and Sawang
    b. 1 operational non-coking coal washery at Piparwar
    c. 2 non-operational non-coking coal washeries at Gidi and
    d. Kargali
IV. 1 central workshop at Barkakana
V. 2 regional workshops at Giridih and Bhurkunda
VI. 3 repair workshops at Jarangdih, Tapin North and Dakra

CENTRAL COALFIELDS LIMITED - THE HISTORICAL MARCH :

CCL had a proud past. As NCDC, it heralded the beginning of nationalization of coal mines in India. National Coal Development Corporation Ltd. (NCDC) was set up in October, 1956 as Government-owned Company in pursuance of the Industrial Policy Resolutions of 1948 and 1956 of the Government of India. It was started with a nucleus of 11 old state collieries (owned by the Railways) having a total annual production of 2.9 million tonnes of coal. Until the formation of NCDC, coal mining in India was largely confined to the Raniganj coal belt in West Bengal and the Jharia coalfields in Bihar (now in Jharkhand), besides a few other areas in Bihar (now in Jharkhand) and a part of Madhya Pradesh (now Chattishgarh also) and Orissa.

From its very beginning, NCDC addressed itself to the task of increasing coa production and developing new coal resources in the outlying areas, besides introducing modern and scientific techniques of coal mining.

In the Second Five Year Plan (1956-1961) NCDC was called upon to increase its production from new collieries, to be opened mainly in areas away from the already developed Raniganj and Jharia coalfields. Eight new collieries were opened during this period and the production increased to 8.05 million tonnes by the end of Second Plan. Third Five Year Plan (1961-1966), though the Corporation had built up a much larger production capacity, it could not be utilized due to a sluggish domestic coal market.



Production had, therefore, to be pegged down and the development of sevejal collieries undertaken from the early part of the Plan period, had to be suspended. By this time, the contribution of NCDC to the nation's coal production (67.72 million tones) increased to around 9.6 million tonnes, With gradual rise in the demand of coal due to commissioning of new power plants and development of other coal-based industries during Fourth Five Year Plan (1969-1974), NCDC's

ROAD

Roads in India are an important mode of transport in India. India has a network of over 6,617,100 km of roads. As of Dec 2024, India has the largest road network in the world.[1] At (1.94 km, 1.21 mi) of roads per square kilometre of land, the quantitative density of India's road network is equal to that of Hong Kong, and substantially higher than the United States (0.71 km, 0.44 mi), China (0.54 km, 0.34 mi), Brazil (0.23 km, 0.14 mi) and Russia (0.09 km, 0.056 mi).[1] Adjusted for its large population, India has approximately 5.13 kilometres (3.19 mi) of roads per 1,000 people, which is much lower than United States 20.5 kilometres (12.7 mi) but higher than that of China 3.6 kilometres (2.2 mi). India's road network carries over 71% of its freight and about 85% of passenger traffic.

Since the 1990s, major efforts have been underway to modernize the country's road infrastructure. As of 31 March 2020, 70.00% of Indian roads were paved. As of 31 December 2023, India had completed and placed into use over 35,000 kilometres (22,000 mi) of four or more lane highways connecting many of its major manufacturing, commercial and cultural centres. According to the Ministry of Road Transport and Highways, as of March 2021, India had about 151,019 kilometres (93,839 mi) of national highways and expressways, plus another 186,528 kilometres (115,903 mi) of state highways. Major projects are being implemented under the Bharatmala, a Government of India initiative. Private builders and highway operators are also implementing major projects.



Delhi-Meerut Expressway, one of the widest express



way in India with 14 lanesRed Road, a major 6-lane highroad in Kolkata

Organization

The Indian road network is administered by various government authorities, given India's federal form of government. The following table shows the total length of India's road network by type of road and administering authority as of 31 March 2020.

Category	Managing Authority	Length (km)	Length percentages
National highways	Ministry of Road Transport and Highways	151,000[3]	2.19%
State highways	Public works department of state/union territory	186,528	3.00%
District Roads	Public works department of state/union territory	632,154	10.17%
Rural roads	Panchayats and PMGSY	4,535,511	72.97%
Urban roads	Municipal corporations and municipalities	544,683	8.76%
Project roads	Various government departments of states/union territories, and SAIL, NMDC and BRO	354,921	5.70%
Total	**Total roadways**	6,404,797 km	**100%**

History



State Highway near Ranchi



Evolution of Indian trade networks. The main map shows the routes since Mughal times, Inset A: shows the major prehistorical cultural currents, B: pre-Mauryan routes, C: Mauryan routes, D: routes c. 1st century CE, and E: the Z-shaped region of developed roads.



A section of the Old Grand Trunk Road, one of the oldest roads in Asia. Originally built by Ashoka and rebuilt by Sher Shah Suri. Kos Minars (right) were used to mark major routes.

The first evidence of road development in the Indian subcontinent can be traced back to approximately around 2800 BC in the ancient cities of Harrapa and Mohenjodaro of the Indus Valley civilization. Ruling emperors and monarchs of ancient and medieval India continued to construct roads to connect the cities. The existing Grand Trunk Road was re-built by the Mauryan Empire, and further rebuilt by subsequent entities such as the Sur Empire, the Mughal Empire and the British Empire.

In the 1830s, the British East India Company started a programme of metalled road construction (a.k.a. gravel road), for both commercial and administrative purposes. The Grand Trunk Road – from Calcutta, through Delhi to Peshawar – was rebuilt at a cost of £1,000 per mile; roads from Bombay to Pune, Bombay to Agra and Bombay to Madras were constructed; and a Public Works Department and the Indian Institute of Technology Roorkee were founded, to train and employ local surveyors, engineers and overseers, to perform the work, and to maintain the roads. This programme resulted in an estimated 2,500 kilometres (1,600 mi) of metalled roads being constructed by the 1850s.

In December 1934, the Indian Roads Congress (IRC) was formed, on the recommendations of the Indian Road Development Committee (Jayakar Committee) of the Government of India. In 1943, they

proposed a twenty-year plan to increase the road network from 350,000 kilometres (220,000 mi) to 532,700 kilometres (331,000 mi) by 1963, to achieve a road density of 16 km per 100 km^2 of land. The construction was to be paid in part through the duty imposed, since 1939, on petrol sales. This became known as the Nagpur Plan. The construction target was achieved in the late 1950s. In 1956, a Highways Act was passed, and a second twenty-year plan proposed for the period 1961–1981, with the ambition of doubling road density to 32 km per 100 km^2. This second plan became known as the Bombay Road Plan.[1]

In 1988, an autonomous entity called the National Highways Authority of India (NHAI) was established by an Act of Parliament and came into existence on 15 June 1989. The Act empowered NHAI to develop, maintain and manage India's road network through National Highways. However, little happened until India introduced widespread economic liberalization in the early 1990s. Since 1995, NHAI has increasingly privatized road network development in India.



Golden Quadrilateral connects the four major Metropolitan Cities of India, viz., Delhi (north), Kolkata (east), Chennai (south) and Mumbai (west).



NH76: Part of India's Golden Quadrilateral highway network

. The flagship project of the NHDP is the Golden Quadrilateral, a total of 5,846 kilometres (3,633 mi) of four-to-six-lane highways connecting the four major cities of Delhi, In 1998, National Highways Development Project (NHDP) was started by the then Prime Minister Atal Bihari VajpayeeMumbai, Chennai and Kolkata. The total cost of the project is ₹300 billion (US$3.5 billion), funded largely by the government's special petroleum product tax revenues and government borrowing. In January 2012, India announced that the four-lane GQ highway network was complete.[13][14]

Another important road project of the NHDP is the 7,142-kilometre (4,438 mi) four-to-six-lane North–South and East–West Corridor, comprising national highways connecting four extreme points of the country. The project aims to connect Srinagar in the north to Kanyakumari in the south (including a spur from Salem to Kanyakumari, via Coimbatore and Kochi), and Silchar in the east to Porbandar in the west. As of 31 October 2016, 90.99% of the project had been completed, 5.47% of the project work is under implementation and 3.52% of the total length is remaining.

As of May 2017, under NHDP, about 28,915 kilometres (17,967 mi) of four-to-six-lane highways have been constructed (including the GQ and N–S/E–W Corridor), while a total of 48,793 kilometres (30,319 mi) of road has been planned to have four-to-six lanes under the NHDP.

The National Highways and Infrastructure Development Corporation Limited (NHIDCL) is a Public Sector Enterprise(PSE) created by

the Ministry of Road Transport and Highways (MoRTH), Government of India in the year 2014 to build highways in technical challenging and high altitude regions of the Northeast India, Uttarakhand, Jammu and Kashmir, Ladakh and the Andaman & Nicobar Islands. It has the task to implement the Special Accelerated Road Development Programme for North Eastern Region (SARDP-NE) in National Highways portion. The SARDP-NE is under implementation in Phases.

- Phase-A: Approved in 2005, it included about 4,099 km length of roads (3,014 km of NH and 1,085 km of State roads). The SARDP-NE Phase 'A' is expected to be completed by 2023–24.
- Phase-B: It covers 3,723 km (2,210 km NHs and 1,513 km of State roads) of road. Phase 'B' of SARDP-NE shall be taken up after completion of Phase 'A'.

Bharatmala is a centrally-sponsored and funded road and highways project of the Government of India, started in 2017, with a target of constructing 83,677 km (51,994 mi) of new highways at an estimated cost of ₹5.35 trillion (US$62 billion). Bharatmala Phase I plans to construct 34,800 kilometres (21,600 mi) of highways (including the remaining projects that were under NHDP) by 2021–22, at an estimated cost of ₹535,000 crore (US$62 billion). In 2021, Asia's longest high speed track, National Automotive Test Track was inaugurated in Indore, which would be used to measure the maximum speed capabilities of high-end cars and other categories of vehicles.

India's rate of road building has accelerated since 2010s. It averaged about 12 kilometres (7.5 mi) per day in 2014–15 and 30 kilometres (19 mi) per day in 2018–19. The country's target is to build 40 kilometres (25 mi) of highways per day.

On July 21, 2021, the Minister of Road Transport and Highways Nitin Gadkari said that India has created a world record of constructing 2.5 kilometres (1.6 mi) of four-lane concrete road in 24 hours and 26

kilometres (16 mi) of single lane bitumen road in just 21 hours as per the highest IRC norms and specifications of the MoRTH to ensure quality control. Also, an average of 36.5 kilometres (22.7 mi) of highways have been constructed every day during 2020–21. As of 2021, 64.5% of all goods in India are moved through the country's road network, 90% of India's total passenger traffic uses the road network to commute and the road network contributes 4.8% to the country's gross domestic product.

In 2023, India's road network became the world's second largest, after the United States. From 2013 to 2014 to 2022 to 2023, the country's road network grew by approximately 59%.In August 2023, the Border Roads Organisation, a statutory body under the Ministry of Defence, began construction on the Likaru-Mig La-Fukch road in Ladakh, which on its completion will be the world's highest motorable road.

Growth of Road Network by Categories (km)

	1950-51	**1960-61**	**1970-71**	**1980-81**	**1990-91**	**2000-01**	**2010-11**	**2015-16**	**2020-21**
National Highways	19,811 (4.95%)	23,798 (4.54%)	23,838 (2.61%)	31,671 (2.13%)	33,650 (1.45%)	57,737 (1.71%)	70,934 (1.52%)	101,011 (1.80%)	**151,000 (2.51%)**
State highways	^	^	56,765 (6.20%)	94,359 (6.35%)	127,311 (5.47%)	132,100 (3.92%)	163,898 (3.50%)	176,166 (3.14%)	**186,528 (3.00%)**
District roads	173,723 (43.44%)	257,125 (49.02%)	276,833 (30.26%)	421,895 (28.40%)	509,435 (21.89%)	736,001 (21.82%)	998,895 (21.36%)	561,940 (10.03%)	**632,154 (10.17%)**

Rural roads	206,408 (51.61%)	197,194 (37.60%)	354,530 (38.75%)	628,865 (42.34%)	1,260,430 (54.16%)	1,972,016 (58.46%)	2,749,804 (58.80%)	3,935,337 (70.23%)	**4,535,511 (72.97%)**
Urban roads	0	46,361 (8.84%)	72,120 (7.88%)	123,120 (8.29%)	186,799 (8.03%)	252,001 (7.47%)	411,679 (8.80%)	509,730 (9.10%)	**544,683 (8.76%)**
Project roads	0	0	130,893 (14.31%)	185,511 (12.49%)	209,737 (9.01%)	223,665 (6.63%)	281,628 (6.02%)	319,109 (5.70%)	**354,921 (5.71%)**
Total	399,942	524,478	914,979	1,485,421	2,327,362	3,373,520	4,676,838	5,603,293	**6,215,797**

Figures in parentheses indicate the percentage of total road length for that fiscal year.

Types of roads

Expressways



Purvanchal Expressway, UP

Main article: Expressways of India

As per NHAI and Indian Roads Congress, expressways are access controlled highways with a divided carriageway, designed for high speed vehicular movement and heavy traffic. Most of the existing

expressways in India are toll roads. Expressways make up approximately 5,579 km (3,467 mi) of India's road network, as of 2024.

National Expressways Authority of India (NEAI) operating under the Ministry of Road Transport and Highways will be in-charge of the construction and maintenance of expressways. The NHAI by Government of India aims to expand the expressway network and plans to add an additional 18,637 km (11,580 mi) of expressways by 2024 apart from existing national highways.

India's first 8-lane wide access-controlled expressway, the Delhi Noida Direct Flyway (DND Flyway), operational in January 2001, is an expressway connecting Delhi and Noida in the states of Delhi and Uttar Pradesh. The Mumbai Pune Expressway, connecting Mumbai and Pune in Maharashtra fully operational in 2002, is India's first 6-lane wide access-controlled tolled expressway. The Yamuna Expressway is a 165 km (103 mi) six-lane controlled-access expressway opened on 9 August 2012. On 21 November 2016, the 302 km (188 mi) six-lane Agra Lucknow Expressway was opened. Under construction as of 2019, the Mumbai–Nagpur Expressway is expected to become the largest expressway in the country. Several expressway projects, such as the Delhi–Mumbai Expressway, Amritsar–Jamnagar Expressway, Surat–Chennai Expressway, Delhi-Jaipur Expressway, Lucknow-Kanpur Expressway are planned/under-construction. Ganga Expressway is approved and expected to be under-construction by end of 2021.

The Trans Harbour bridge is the longest bridge in India and it will be opened on 12 January 2024, after Prime Minister Narendra Modi inaugurates the bridge. It connects Bombay with Navi Mumbai.

National highways

Main article: National highways of India



North-South East-West Corridors



connects Srinagar to Kanyakumari and Porbandar to Silchar

NH75: Part of India's NS and EW Corridor highway network

National highways are highways connecting major cities throughout the country with premium quality and are at-grade roads. National Highways are designated with NH, followed by the highway number. Indian national highways are further classified based on the width of the carriageway of the highway. India has around 150,000 km (93,000 mi) of National Highways as of April 2021 and is expected to reach 200,000 km By 2024 consisting of Top Notch Highways And Expressways. National Highways constituted 2.7% of India's total road network, but carried about 40% of road traffic, as of 2013. In 2016, the government vowed to double the highway length from 96,000 to 2,00,000 km.

The National Highways Authority of India (NHAI) and the National Highways and Infrastructure Development Corporation Limited (NHIDCL) are the authorities responsible for the development, maintenance and management of the National Highways in India. The NHAI has been undertaking developmental activities under the National Highways Development Project (NHDP) in five phases. From 2018, the pending projects under NHDP are expected to be subsumed under Bharatmala. The NHAI is also responsible for implementing other projects on National Highways, primarily road connectivity to major ports in India.

The Golden Quadrilateral and North–South and East–West Corridor were major ongoing highway development projects in India.

National Highway classification (as of 31 March 2020)

Lanes	Length (km)	Length share
Single-lane (3.75 m / 12.3 ft) or Intermediate lane (5.5 m / 18 ft)	29,693	20.49%

National Highway classification (as of 31 March 2020)

Lanes	Length (km)	Length share
Double-lane (7 m / 23 ft without paved kerb) and (7.5 m / 24.6 ft with paved kerb)	72,281	55.05%
Four-lane / Six-lane / Eight-lane	37,058	24.46%
Total	1,39,032	100%



Gujarat State Highway 41

State highways

Main article: State highways in India

State highways are highways connecting major cities throughout a state and are also at-grade roads. They also connect with National Highways or state highways of neighboring states. State Highways are designated with SH, followed by the highway number and preceded by state code. As of 31 March 2020, the total length of state highways was 186,528 kilometres (115,903 mi). As of 31 March 2020, Maharashtra has the largest share of state highways among all

states (22.14%), followed by Karnataka (11.11%), Gujarat (9.76%), Rajasthan (8.62%) and Tamil Nadu (6.67%).

State governments have the authority and responsibility to build state highways. Most of the state highways are developed by state public works departments. Independently of the Bharatmala program, state governments have been implementing a number of state highway projects since 2000. By 2010, state highway projects worth US$1.7 billion had been completed, and projects worth an additional US$11.4 billion were under implementation.

District roads



A district road in Thiruvananthapuram

District Roads in India are approximately 632,154 kilometres (392,802 mi), of which 14.80% of the total length was surfaced.Zila Parishads also have the authority and responsibility to build district roads.

Rural roads





A rural road in Jharkhand A rural road in Chhattisgarh

Rural roads form a substantial portion of the country's road network, forming 72.97% of the total of roads, as of March 2020. As of the same date, the percentage of unsurfaced roads to the total road length was 31%.

For the development of these rural roads, Pradhan Mantri Gram Sadak Yojana (Prime Minister's Rural Roads Scheme) was launched in December 2000 by the Indian government to provide connectivity to isolated rural habitations. The scheme envisions that these roads will be constructed and maintained by the village panchayats. In some parts of India, the government has attempted to manage the programme directly as a local social spending program.

In other parts of India, the *Pradhan Mantri Gram Sadak Yojana* and a sister program named *Bharat Nirman* (Build India) have privatized the rural road construction projects and deployed contractors. The effort has aimed to build all-season single-lane asphalted roads to connect India's rural and remote areas. A significant portion of funding for these projects has come from the World Bank and the Asian Development Bank.

Growth of Rural Road Network (km)				
	Length 2001	**Length 2011**	**Length 2021**	
Total rural roads	2.7 million	3.1 million	4.5 million	
Paved unmaintained rural roads	0.5 million			
Unpaved rural roads	2.2 million	1.9 million		
Paved maintained rural roads		728,871		
New rural roads		322,900	82,743	1,500,000

Border roads

See also: India-China Border Roads and Northeast connectivity projects

Border Roads are the roads constructed along the northern and northeastern borders of the country. These roads are constructed and maintained by Border Roads Organisation (BRO) which was set up in 1960 by the government of India. BRO is regarded as a symbol of nation building, national integration and an inseparable component in maintaining the security of the country.

Congestion

India's intra-city vehicle speed is among the lowest in the world. As per a study by Ola Cabs, in 2017, the average traffic speed in Delhi was 25 km/h (16 mph). Amongst other major cities, the average traffic speed in Chennai was 18.9 km/h (11.7 mph), in Mumbai was 20.7 km/h (12.9 mph), in Kolkata was 19.2 km/h (11.9 mph), in Hyderabad was 18.5 km/h (11.5 mph), and in Bengaluru was 17.2 km/h (10.7 mph).

Fatalities

The World Health Organization's compilation of road network safety data for major economies found India to have the highest number of road fatalities in the world, with 299,091 deaths caused by road accidents in 2016. Also, fatalities per 100,000 population stay among the highest, at 22.6. Of total fatalities, 40% were of riders of 2 or 3 wheelers, 18% were of drivers and passengers of 4-wheeled cars and light vehicles, 18% were of drivers and passengers of buses and heavy trucks, 10% of pedestrians, 2% of cyclists and 13% of other.

PAVEMENTS



What are Pavements?

Pavements are the upper layers of a roadway structure that provide a smooth and durable surface for vehicles and pedestrians. Their primary role is to distribute traffic loads to the underlying layers of the road and prevent deformation or failure of the subgrade (natural soil). Pavements are essential for efficient and safe transportation systems and come in various types, each suited for specific needs and conditions.

They serve three key purposes:

1. To support and distribute the load from traffic.
2. To resist adverse weather conditions (heat, rain, snow).
3. To provide comfort, durability, and safety for road users.

Requirements of a Good Pavement

A well-designed pavement must meet several functional and structural requirements to ensure long-lasting and safe performance. These include:

- **Structural Strength:** It should support and distribute traffic loads without causing excessive deformation or rutting.
- **Durability:** The pavement must withstand environmental factors like rain, heat, frost, and oxidation.
- **Skid Resistance:** The surface should provide adequate friction to prevent skidding and ensure vehicle control.
- **Smoothness:** A good pavement offers a smooth and comfortable ride, minimizing wear and tear on vehicles.
- **Drainage:** Proper drainage must be ensured to prevent water stagnation, which weakens pavement layers.
- **Low Maintenance:** A good pavement should require minimal repairs and maintenance over its lifecycle.
- **Economical:** It should offer long service life and value for money in terms of construction and upkeep.

Types of Pavements

Pavements are mainly classified based on the materials used and the way they distribute loads:

1. Flexible Pavements

These are made using bituminous (asphalt) materials and consist of multiple layers. Load distribution happens through grain-to-grain contact, and the deformation of layers is gradual and flexible.

2. Rigid Pavements

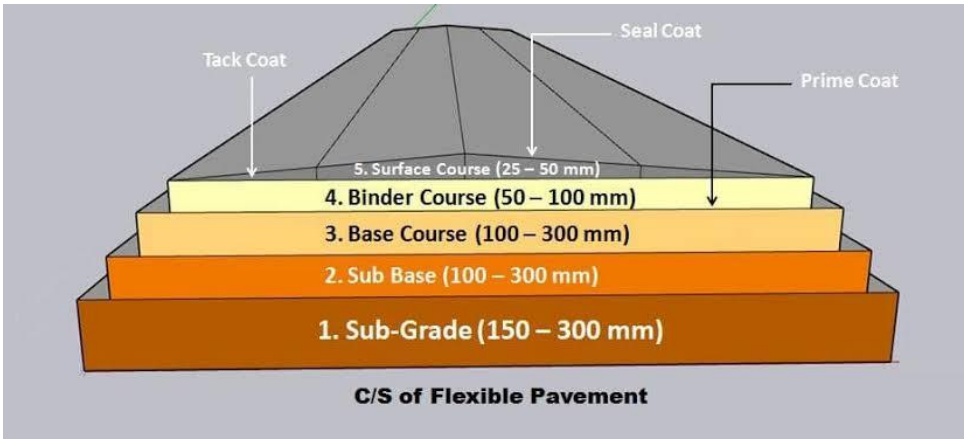
Made of cement concrete, rigid pavements behave like a beam and distribute loads over a wide area through slab action. These pavements are stiffer and have a longer service life.

3. Semi-Rigid Pavements



These are composite pavements that combine properties of both flexible and rigid pavements. They include treated bases such as lime or cement-stabilized layers, providing higher strength than traditional flexible pavements.

FLEXIBLE PAVEMENTS



Types of Flexible Pavements

Flexible pavements can be divided into several categories depending on the materials and construction techniques:

1. **Conventional Flexible Pavements:**
    - Consist of surface course (asphalt), base course, sub-base, and subgrade.
    - Common in urban and rural roads.
2. **Stabilized Base Pavements:**
    - Use soil or aggregates mixed with stabilizing agents like lime, cement, or bitumen in the base or sub-base layers.
    - Offers better strength and moisture resistance.
3. **Full-Depth Asphalt Pavements:**
    - All layers above subgrade are asphalt.
    - Used in areas with high traffic load like airports and industrial roads.
4. **Composite Pavements:**

- A bituminous overlay over an existing concrete (rigid) pavement.
- Combines the smooth riding quality of asphalt with the strength of concrete.

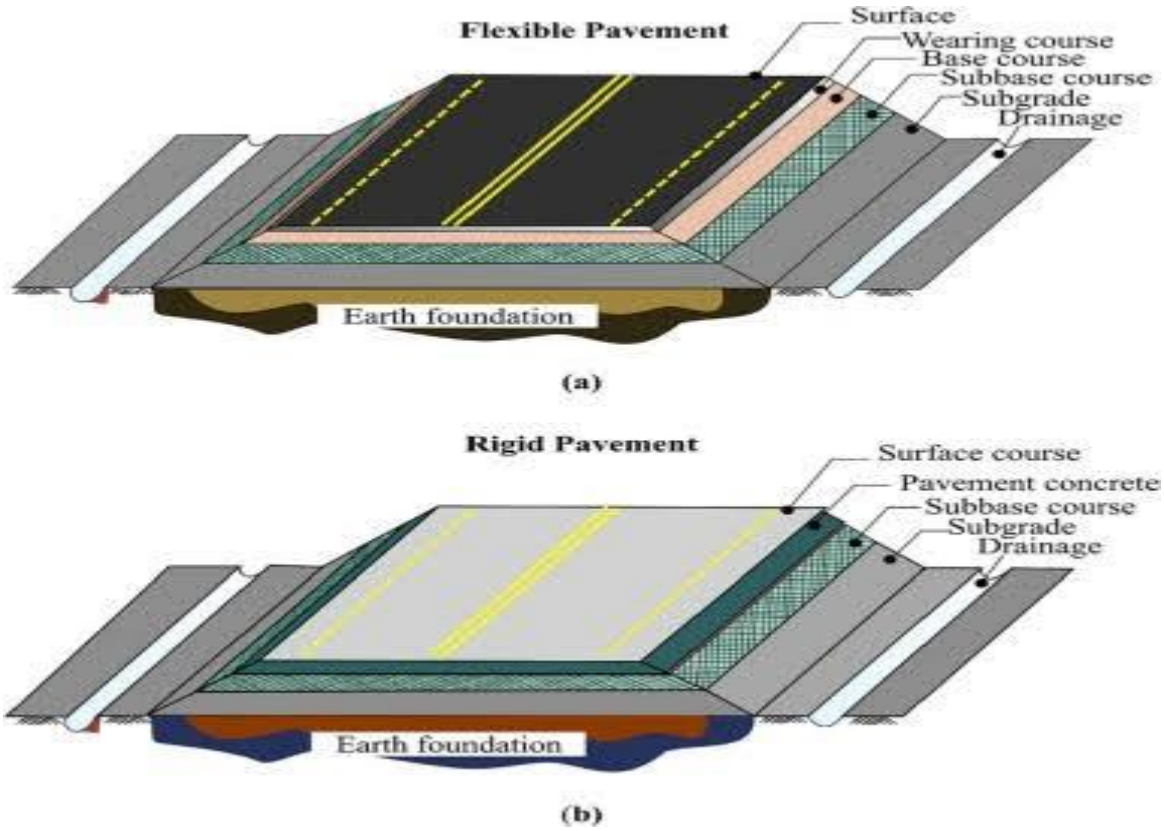
Layers of Flexible Pavement / Road Construction Layers

Flexible pavements consist of several structural layers, each serving a specific function:

1. **Subgrade:**
   - The natural soil layer, compacted to bear the load from upper layers.
   - Acts as a foundation; poor subgrade may lead to pavement failure.
2. **Sub-Base Course:**
   - Laid over the subgrade, made of lower quality materials.
   - Provides additional support and drainage, especially in low-traffic roads.
3. **Base Course:**
   - High-quality crushed aggregates or granular materials.
   - Provides major structural capacity and distributes traffic loads.
4. **Binder Course:**
   - An intermediate asphalt layer between the base and the surface.
   - Reduces the effect of load stresses and adds durability.
5. **Wearing Course (Surface Course):**

- Top layer made of high-quality bitumen mix.
- Provides skid resistance, waterproofing, and a smooth ride.

Rigid Pavement



Rigid pavements are constructed using Portland cement concrete and are characterized by high stiffness and durability. They behave like a single slab or beam, transmitting loads through a wide area. These pavements have high initial construction costs but offer long-term performance with lower maintenance.

They are suitable for:

- High-speed highways
- Airport runways
- Heavy-load zones like industrial roads

Advantages:

- Long service life (20–30 years)
- Lower maintenance frequency
- Better performance in areas with high water table or black cotton soil

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Types of Rigid Pavements

Rigid pavements are categorized based on the type and amount of reinforcement used:

1. **Jointed Plain Concrete Pavement (JPCP):**
    - Plain concrete slab with contraction joints to control cracking.
    - No reinforcement.
    - Common on highways and city roads.
2. **Jointed Reinforced Concrete Pavement (JRCP):**
    - Includes both joints and steel reinforcement.
    - Controls crack width and increases slab length.
    - Used where slab dimensions are large.
3. **Continuously Reinforced Concrete Pavement (CRCP):**
    - No joints; continuous reinforcement is provided.

- Designed to form fine, tightly held cracks throughout the pavement.
- Suitable for very high traffic loads.

Flexible vs. Rigid Pavement (Comparison Table)

Feature	Flexible Pavement	Rigid Pavement
Material Used	Bituminous (asphalt) layers	Cement concrete
Load Distribution	Layer by layer (grain to grain)	Through slab action
Initial Cost	Lower	Higher
Maintenance	Frequent	Rare and less frequent
Service Life	10–15 years	20–30 years or more
Surface Flexibility	Flexible – deforms slightly	Rigid – little to no deformation
Construction Time	Quicker	Slower due to curing time
Suitable For	Low to medium traffic roads	High traffic and industrial roads

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Conclusion

Pavements are critical components of transportation infrastructure. The choice between flexible and rigid pavements depends on factors like expected traffic load, climate, subgrade conditions, and budget.

Advances in materials, such as recycled plastics and smart sensors, are transforming pavement technology. A good pavement ensures safe travel, low maintenance costs, and a long service life.

PQC



1. What is PQC (Pavement Quality Concrete)?

PQC refers to a special grade of concrete used for rigid pavement construction, particularly for roads that require high strength and durability. It is typically used in **concrete highways, expressways, airport runways**, and **industrial pavements** where the load-bearing requirements are higher than ordinary roads.

Key Characteristics:

- Made with **M40 or higher grade concrete**
- High **compressive and flexural strength**
- Designed for **long service life** and low maintenance
- Placed using **slip-form or fixed-form paving equipment**

2. PQC Road – Definition

A **PQC Road** is a concrete pavement constructed using Pavement Quality Concrete. These roads have a high load-carrying capacity, making them suitable for heavy traffic areas such as:

- **National highways**
- **Airport taxiways and runways**
- **Container terminals**
- **Toll plazas and industrial zones**

3. PQC Construction Method

Steps in PQC Road Construction:

1. **Preparation of Subgrade and Sub-base**
    - Subgrade is compacted, leveled, and graded.
    - A **Dry Lean Concrete (DLC)** layer (typically 150 mm thick) is placed as a sub-base.
2. **Placement of PQC**
    - PQC is placed over the DLC using either:
        - **Slip-form paver** (automated machine), or
        - **Fixed-form paver** (manual using side forms)
    - Concrete is spread, vibrated, and compacted.
3. **Joint Cutting**
    - **Contraction and expansion joints** are created within 4–6 hours using diamond saw cutters.
4. **Curing**
    - PQC is cured using wet gunny bags, continuous sprinkling, or curing compounds for a minimum of 14 days.

4. PQC Construction Materials

Key Materials Used in PQC:

1. **Cement**: Ordinary Portland Cement (OPC) – Grade 43 or 53
2. **Fine Aggregates**: Well-graded river sand or crushed sand
3. **Coarse Aggregates**: Crushed stone with size ranges 20 mm and 10 mm
4. **Water**: Potable, free from harmful salts
5. **Admixtures**: Superplasticizers (for better workability)
6. **Steel Dowel Bars & Tie Bars**: For load transfer at joints

5. PQC Construction Materials – Important Notes

- **Cement content** is usually kept high (around 360–400 kg/m³) to achieve high strength.
- **Water-cement ratio** should be low (0.35 to 0.40) for better durability.
- **Air entrainment** is avoided in PQC unless required for freeze-thaw conditions.
- **Maximum aggregate size** should not exceed 31.5 mm.
- **Superplasticizers** improve workability without increasing water content.

6. Construction of PQC

Detailed Construction Steps:

1. **Sub-base Preparation:**
    - A **Dry Lean Concrete (DLC)** layer is laid and compacted.
2. **Formwork (for fixed-form paving):**
    - Side forms are fixed to guide concrete placement.
3. **Concrete Placement:**
    - PQC is poured in panels using transit mixers or dumpers.
4. **Vibration & Compaction:**
    - Internal and surface vibrators are used to remove air voids.
5. **Finishing:**
    - Surface is leveled and textured using brooming to prevent skidding.
6. **Curing:**
    - Curing begins after joint cutting and continues for 14 days.

7. Contraction Joints

- **Purpose:** To control random cracking by providing weak planes where concrete can shrink.
- **Spacing:** Usually 4.5 m to 5 m apart.
- **Depth:** 1/3rd of slab thickness.
- **Method:** Created using saw-cutting after initial set (4–6 hours).
- **Dowel bars** are provided across the joint for load transfer.

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8. Expansion Joints

- **Purpose:** To accommodate expansion due to temperature changes.
- **Location:** Usually at structures (bridges, culverts, ends of pavements).
- **Materials Used:** Compressible fillers like bitumen boards.
- **Spacing:** Typically 50–100 m intervals or as specified.
- **Dowels** are placed with one end capped to allow movement.

9. Concrete Mix Design Requirements for PQC

Typical PQC Mix Design Targets:

- **Grade:** M40 or higher (compressive strength ≥ 40 MPa in 28 days)
- **Flexural Strength:** ≥ 4.5 MPa
- **Water-Cement Ratio:** 0.35–0.40
- **Slump:** Very low (0 to 25 mm), as PQC is laid in dry, stiff consistency
- **Aggregate Proportion:** 60–70% of total mix by volume
- **Admixtures:** Superplasticizers for workability; no air-entraining agents unless specified
- **Cement Content:** Around 370–400 kg/m³

Standards Followed:

- **IRC:15**, **IRC:58**, and **IS 10262** for mix design
- Quality control is essential through lab and field tests (slump, cube, flexural strength, thickness)

1. Introduction to WBM Roads



Water Bound Macadam (WBM) roads are a type of low-cost road construction where **coarse aggregates** are bound together using **screening materials** and **binding agents**, and are compacted by rolling while water is sprinkled.

It was developed by **John Loudon McAdam** in the early 19th century and remains widely used in rural, low-traffic areas and as a base course for bituminous roads.

2. Types of Macadam Roads

1. **Water Bound Macadam (WBM):**
   Uses water and binding material for compacting crushed stones.
2. **Traffic Bound Macadam:**
   Compaction is achieved through traffic movement without water or binding materials.
3. **Bituminous Bound Macadam (BBM):**
   Uses bitumen as the binding agent; commonly used as a base course.
4. **Cement Bound Macadam (CBM):**
   A cement slurry is used to bind aggregates.

3. WBM Road Cross Section

A typical WBM road cross section includes the following:

- **Subgrade (natural soil)**
- **Sub-base course (optional)**
- **Base course (WBM layers)**
- **Surface course (optional in high-volume roads)**
- **Shoulders (for lateral support and drainage)**

Cross Slope (Camber):
2.5% to 3% to allow water runoff

Total Width:
Typically 3.75 m for a single lane

4. Macadam Road Construction Method

The method includes preparation of the subgrade, placing and compacting layers of **coarse aggregates**, **screenings**, **binding materials**, and **sprinkling water** during each stage.

5. Materials Required for WBM Roads

1. **Coarse Aggregates**
2. **Screenings**
3. **Binding Materials**
4. **Water**

6. Coarse Aggregates

- **Material:** Crushed stone, granite, basalt
- **Size:** 90–40 mm (for base), 63–40 mm, 50–20 mm (for top layers)
- **Los Angeles Abrasion Value:** ≤ 40%
- **Impact Value:** ≤ 30%
- Should be clean, angular, and durable

7. Screenings

- **Purpose:** Fills voids between coarse aggregates
- **Material:** Stone dust, kankar, or crushed sand
- **Grading:** Depends on the layer thickness and type

Example:

- Grading A: 10 mm down stone dust

- Grading B: 6 mm down screeninG

8. Binding Material

- Used to **cement the particles together** after rolling and sprinkling
- **Plasticity Index (PI):** Between 4 to 9
- **Materials:** Moorum, clay, or soil mixed with fine sand
- Should not be expansive or highly plastic

9. Construction Procedure of WBM Road

A. Foundation Preparation for Receiving WBM Course

- **Subgrade is excavated** and compacted
- If needed, a **sub-base layer** (e.g., sand, gravel, or GSB) is added
- The surface should be well-drained and properly cambered

B. Lateral Confinement of Aggregates

- **Shoulders** are built first to confine coarse aggregates
- Shoulders are about 25–30 cm higher than the adjacent layers during rolling

C. Spreading Coarse Aggregates

- Coarse aggregates are uniformly spread using mechanical spreaders or manually
- Layer thickness: **75 mm compacted**
- Layer is **checked with templates** for uniformity

D. Rolling

- Rolling begins from the edges and progresses to the center
- **Roller Type:** 6–10 ton smooth wheeled roller
- Rolling is done until aggregates are well interlocked and no visible movement occurs

E. Application of Screenings

- Screenings are applied after partial rolling
- They are broomed into voids left between aggregates
- Water is sprinkled during brooming to assist binding

F. Sprinkling and Grouting

- **Water is sprinkled** over the surface
- The voids are filled and compacted fully
- Rolling continues until slurry forms on the surface and all voids are filled

G. Application of Binding Material

- After initial rolling, **binding material is added**
- Water is sprinkled, and final rolling is carried out
- This forms a stable surface

H. Setting and Drying

- Road is allowed to **set and dry for 24–48 hours**
- Traffic is not allowed during this period
- If necessary, the surface is lightly watered and re-rolled

10. Maintenance of WBM Roads

- **Regular grading** to fix ruts and potholes
- **Reapplication of screenings or binding material**
- **Dust control** through sprinkling
- **Drainage maintenance** to avoid water stagnation
- **Edge repair** for lateral support

11. Advantages of WBM Roads

- **Low Cost**: Economical to construct and maintain
- **Locally Available Materials**: No need for imported bitumen or cement
- **Simple Construction**: No heavy machinery needed
- **Flexibility**: Can be constructed in remote areas with limited resources
- **Base for Bituminous Roads**: Often used as a base layer

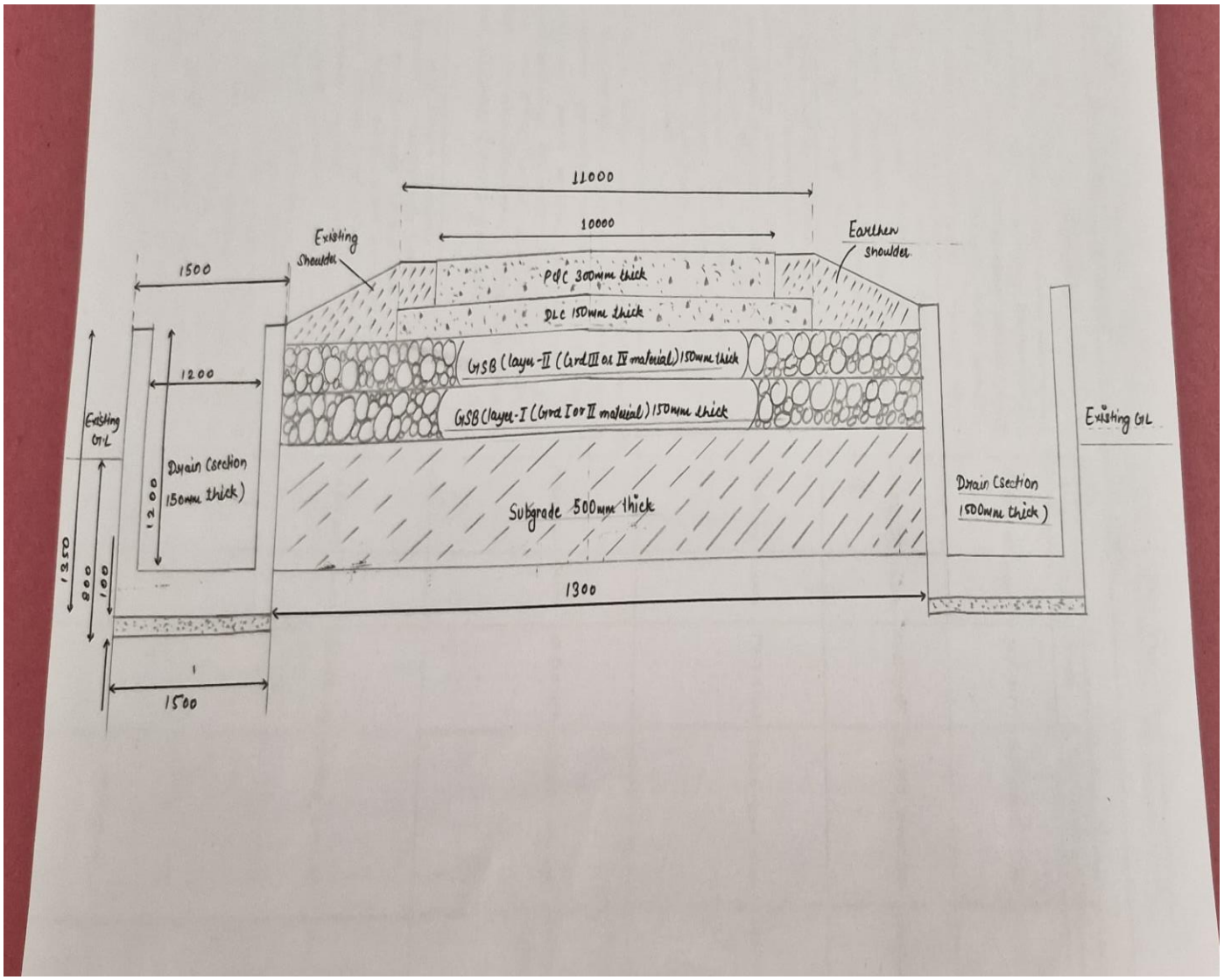
12. Disadvantages of WBM Roads

- **Dust Generation**: Major issue in dry areas
- **High Maintenance**: Frequent repairs needed due to loose stones and ruts

- **Not Suitable for Heavy Traffic**: Surface breaks down under repeated loads
- **Water Sensitivity**: Loses strength in wet conditions
- **Uneven Surface**: Compared to bitumen roads

Conclusion

WBM roads are ideal for **rural and low-volume roads** due to their simplicity and low cost. However, proper materials, construction, and maintenance are critical to ensuring their effectiveness and lifespan.



Bill of Quantity

Name of Work: Estimate for construction of PQC Road from zero point to coal stock no. 53 under Amaypali project of A-C Area.

PART-A (PQC Road)

Sl. no.	Item of Work	Qty	Unit	Rate	Amount (Rs)	Remark
1.01	Removal of unserviceable soil including excavation, Loading and disposal upto 1000 meters lead but excluding replacement by suitable soil which shall be paid separately as per clause 305.	3900.00	cum	37.00	144,300.00	NH3.11
1.02	Construction of embankment with approved material obtained from borrow pits with all lifts and leads, transporting to site, spreading grading to required slope and compacting to meet requirement of table 300.2	6500.00	cum	178.00	1,157,000.00	NH3.16
1.03	Granular sub-Base with coarse graded material (Table-400.2) (Construction of granular sub-base by providing coarse graded material, spreading in uniform layers with motor grader on prepared surface, mixing by mix in place method with rotavator at OMC and compacting with vibratory roller to achieve the desired density, complete as per clause 401) for grading-I Material.	1950.00	cum	1974.00	3849,300.00	NH4.2 (I)
1.04	Granular sub-Base with coarse Graded material (Table 400.2) (construction of granular sub base by providing coarse graded material, spreading in uniform layers with motor grader on prepared surface, mixing by mix in place method with rotavator at OMC, and compacting with vibratory roller to achieve the desired density complete as per clause (401) for grading-III material.	1950.00	cum	1258.00	2,452,100.00	NH4.2 (III)
1.05	Construction of dry lean cement concrete sub-base over a prepared subgrade with coarse and fine aggregate conforming to IS:383, the size of coarse aggregate not exceeding 25mm aggregate cement ratio not to be exceed 15:1, aggregate gradation after bleeding to be as per specification, cement content to be determined during trial length construction, concrete strength not be less than 10 MPa at 7 days, mixed in a batching plant, transported to site for all leads & lifts, laid with a mechanical paver, compacting with 8-10 tonne vibratory roller finishing and curing etc. complete as per direction of Engineer-incharge.	1650.00	cum	3542.00	5,844,300.00	NH6.1

Sl No.	Item of work	Qty.	Unit	Rate	Amount (Rs)	Remark
1.06	Cement concrete Pavement (construction of un-reinforced dowel jointed, plain cement concrete pavement over a prepared sub base with 43 grade @ 400kg per cum. coarse and fine aggregate conforming to IS 383, maximum size of coarse aggregate not exceeding 25mm, mixed in a batching and mixing plant as per approved mix design, transported to site, laid with a fixed form or slip form paver, spread, compacted and finished in a continous separation including provisional of contraction and longitudinal joints, joint filler, separation membrane, sealant primer, joint sealant, debonding skip, dowel bar, tie rod admixture as approved, curing compound, finishing to lines and grade as per drawing.	3000.00	cum	6404.00	19,121,000.00	NH-6.2
1.07	construction of sub-grade and earthen shoulders with approved material obtained from borrow pits with all lifts & leads transporting to site, spreading, grading to required slope and compacted to meet requirement table No. 300-2.	960.00	cum	189	171,160.00	3.18
	PART-B (Drain) (RCC Drain)					
2.01	Earthwork in excavation of foundation of structure as per drawing and technical specification, including setting out, construction of shoring and bracing, removal of stumps & other deleterious matter dressing of sides and bottom, backfilling the excavation earth to the extent required and utilising the remaining earth locally for road work.	2400.00	cum	41.00	98,400.00	3.13(b)
2.02	Plain cement concrete 1:3:6 nominal mix in foundation with crushed stone aggregate 40mm nominal size mechanically mixed placed in foundation and compacted by vibration including curing for 14 days.	300.00	cum	4288.00	1,286,400.00	12.4
2.03	Plain / Reinforced cement concrete in open foundation complete as per drawing and technical specification.	1170.00	cum	6298.00	7,368,660.00	12.8(c)
2.04	Supplying, fitting and placing un-coated HYSD bar R/F in foundation complete as per drawing and Technical specification	106.8835	MT	74667.00	7,980,668.20	12.40
					16,734,128.00	

PART-C Guard Wall (RCC Guard Wall)

Sl No.	Item of Work	Qty	Unit	Rate	Amount (Rs)	Remark.
	Earthwork in excavation of foundation of structures as per drawings and technical specifications, including setting out, construction out shoring and bracing, removal of stumps and other deleterious matter, dressing of side and bottom, backfilling the excavation earth to the extent required and utilising the remaining earth locally for road work.	2700.00	cum	41.00	110,700.00	3.13(b)
	Sand filling in foundation Trenches as per drawing & Technical specification.	180.00	cum	396.00	71,280.00	12.3
	Plain cement concrete 1:3:6 nominal mix in foundation with crushed stone aggregate 40mm nominal size mechanically mixed, placed in foundation and compacted by vibration including curing for 14 days.	360.00	cum	4288.00	1,543,680.00	12.4
	Plain /Reinforcement Cement Concrete in open foundation complete as per drawing and technical specification.	2722.50	cum	6298.00	17,146,305.00	12.8(c)
	Supplying, fitting and placing un-coated HYSD bar reinforcement in foundation complete as per drawing and Technical specification.	137.224	MT	74667.00	10,246,104.41	12.40

Total (A+B+C)	29,118,069.41
Add 18% GST	78,682,29761
Total (A+B+C).	14,162,813.53
	92,545,111.18

SN	Item Description	Unit	nos	nos	Length	Width	H/D	Quantity	Remark.
1.05	Construction of Dry lean concrete cement sub base over a prepared subgrade with coarse and fine aggregate conforming IS383 the size the coarse aggregate not exceeding 25mm aggregate cement ratio not to be exceed 15:1, aggregate gradation after blending to be as per specifications, cement content not to be less than 150 kg/cum optimum moisture content to be determined allowing field length construction, concrete strength not to be less than 10 mPa at 7 days. mixed in all batching plant, transported to site for all leads & lifts, laid with the mechanical paver, compacting with 8-10 metric tonnes vibratory roller, finishing and curing etc. complete as per direction of Engineer Incharge.	cum		1	1650.00	11.00	0.15	1650	NH6.1
1.06	Cement Concrete Pavement (construction of un-reinforced, dowel jointed, plain cement concrete pavement over a prepared sub-base with 43 grade cement @ 400kg per cum, coarse and fine aggregate conforming to IS 383, maximum size of coarse aggregate not exceeding 25mm, mixed in batching and mixing plant as per approved mix design, transported to site, laid with a fixed form or slip paver, spread, compacted as finished in a continous operation including provision of contracting expansion, construction and longitudinal joints, joint filler, separation membrane, sealant primer, joint sealant, debonding strip, dowel bar, tie-rod, admixture as approved curing compound, finishing to lines and grades as per drawing).	cum		1	1000	10.00	0.30	3000.00	NH6.2
1.07	Construction of sub-grade and earthen shoulders with approved material obtained from borrow pits with all lifts & leads transporting to site, spreading, grading to required slope and compacted to meet requirement of table No. 300.2.	cum		2	1000.00	0.5x0.3+0.15x1+(0.5x0.3x1)		900.00	NH6.2

Calculation Sheet

PART-B Drain (RCC Drain)

Total length of Drain = 1000.00 mtr x 2 sides

SN	Item Description	Unit	nos	nos	Length	Width	H/D	Qty	Remarks
01	Earthwork in excavation of foundation of structure as per drawing and technical specification, including setting out, construction of shoring and bracing, removal of stumps and other deleterious matter, dressing of side and bottoms, backfilling the excavation earth to the extent required and utilising the remaining earth locally for road work.	cum		2	1000.00	1.50	0.80	2400.00	3.13 (b)
02	Plain cement concrete 1:3:6 nominal mix in foundation with crushed stone aggregate 40 mm nominal size mechanically mixed, placed in foundation and compacted by vibration including curing for 14 days	cum		2	1000.00	1.50	0.10	300.00	12.3
03	Plain / Reinforced cement concrete in open foundation complete as per Drawing and Technical specifications.								12.8 (c)
	Bottom	cum		2	1000.00	1.50	0.150	450.00	
	Side wall	cum	2	2	1000.00	0.150	1.20	720.00	
			Total					1170.00	
04	Supplying, fitting and placing un-coated HYSD bar reinforcement in foundation complete as per drawing and technical specifications								12.40
	Bottom 8mm	Kg	2	10	1000.00	39Kg/m		7800.00	
	Bottom 8 mm	Kg	2	10	1000.00	39Kg/m		7800.00	
	Bottom 10mm	Kg	4	6667	1.50	62Kg/m		24801.99	
	Side wall 8mm	Kg	4	14	1000.00	39Kg/m		21840.00	
	Main bar 10mm	Kg	8	6667	1.35	62Kg/m		44642.23	
				Total				106883.47	

Calculation Sheet

PART-B Drain (RCC Drain)

Total length of Drain = 1000.00 mtr x 2 sides

SN	Item Description	Unit	nos	nos	Length	Width	H/D	Qty	Remarks
01	Earthwork in excavation of foundation of structure as per drawing and technical specification, including setting out, construction of shoring and bracing, removal of stumps and other deleterious matter, dressing of side and bottoms, backfilling the excavation earth to the extent required and utilising the remaining earth locally for road work.	cum		2	1000.00	1.50	0.80	2400.00	3.13 (b)
02	Plain cement concrete 1:3:6 nominal mix in foundation with crushed stone aggregate 40 mm nominal size mechanically mixed, placed in foundation and compacted by vibration including curing for 14 days	cum		2	1000.00	1.50	0.10	300.00	12.3
03	Plain / Reinforced cement concrete in open foundation complete as per Drawing and Technical specifications								12.8 (c)
	Bottom	cum		2	1000.00	1.50	0.150	450.00	
	Side wall	cum	2	2	1000.00	0.150	1.20	720.00	
			Total					1170.00	
04	Supplying, fitting and placing un-coated HYSD bar reinforcement in foundation complete as per drawing and technical specifications								12.40
	Bottom 8 mm	kg	2	10	1000.00	39 kg/m		7800.00	
	Bottom 8 mm	kg	2	10	1000.00	39 kg/m		7800.00	
	Bottom 10 mm	kg	4	6667	1.50	62 kg/m		24801.99	
	Side wall 8 mm	kg	4	14	1000.00	39 kg/m		21840.00	
	Main bar 10 mm	kg	8	6667	1.35	62 kg/m		44642.23	
			Total					106883.47	

Calculation Sheet

PART-C Guard wall (RCC Guard Wall)

Total length of Guard wall = 1000 mtr.

SN	Item Description	Unit	nos	nos	Length	Width	H / D	Quantity	Remarks
01	Earthwork in excavation of foundation of structures as per drawing and Technical specification, including setting out, construction of shoring and bracing, removal of stumps and other deleterious matter, dressing of sides and bottom, backfilling the excavation earth to the extent required and utilising the remaining earth locally for road work.	cum		2	1000.00	1.80	0.75	2700.00	3.13 (b)
02	Sand filling in foundation Trenches as per Drawing & Technical specification.	cum		2	1000.00	1.80	0.05	180.00	12.4
03	Plain cement concrete 1:3:6 nominal mix in foundation with crushed stone aggregate 40mm nominal size mechanically mixed, placed in foundation and compacted by vibration including curing for 14 days.	cum		2	1000.00	1.80	0.10	360.00	12.3
04	Plain / Reinforcement cement concrete in open foundation complete as per Drawing and Technical specifications								
	Bottom	cum		2	1000.00	1.50	0.60	1800.00	
	Wall	cum		2	1000.00	(4.5+3)/2	1.65	1237.50	
	Deduction	cum	2	0.5	1000.00	0.30	0.60	-180.00	
		cum	2	0.5	1000.00	0.30	0.45	-135.00	
		Total						2722.50	
05	Supplying, fitting and placing uncoated HYSD Bar Reinforcement in foundation complete as per drawing and Technical specifications								12.40
	Vertical Rod 12mm	kg	2	8000	2.55	89 kg/m		36312.00	
	Bottom Rod 12mm	kg	2	8000	2.90	89 kg/m		41296.00	
	Vertical Rod 10mm	kg	2	8000	2.55	89 kg/m		25296.00	
	Diskibution Rod 8mm	kg	2	44.0	1000.00	39 kg/m		34320.00	
		Total						137224.00	

Scheme cum estimate for "Construction of Approach cum Total Coal Transportation Road (from HMB More to Weighbridge, Pit office and Substation Premises for

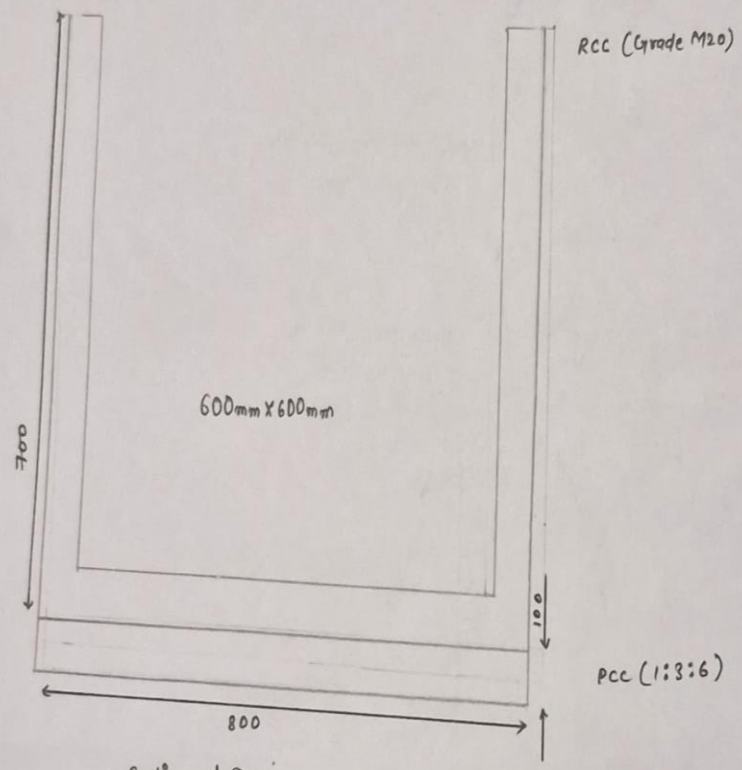
Part - A (Road Work)

Sl No.	Item Description	Unit	NO.	NO	L	B	H	Qty.	Rate as per CCL SoR	Amount	Remarks
1.	Scarifying Existing Bituminous Surface to a depth of 50 mm by Mechanical Means. Scarifying the Existing bituminous surface to a depth of 50 mm and disposal of scarified material with its all lifts and lead upto 1000 metres	Sq m			850.00	5.50		4675.00	7.50	35,062.50	3.15 (Road works)
2.	Excavation for roadwork in soil with hydraulic excavator of 0.9 cum bucket capacity including cutting and loading in tippers, trimming bottom and side slopes, in accordance with requirement of lines, grades & cross sections, and transporting to the embankment location within all lifts and lead upto 1000m (excluding Royalty @ Rs. 24.00 per cum but excluding watering, rolling and compaction).	cum									
					850.00	8.50	0.30	2167.50			The width of road in 8.50m in a Stretch of 850m
					370.00	11.00	0.30	1221.00			
					600.00	11.00	0.30	1980.00			320+110+60+60+50 = 600m
								5368.50	79.00	424,111.50	
3.	Construction of embankment with Material obtained from Borrowpits Construction of embankment with approved material obtained from borrowpits with all lifts and leads, transporting to site, spreading, grading to required slope and compacting to meet requirement of table 300-2.										3.16 (i) (Road works)
		cum		2	850.00	0.894	4.47	6796.75	139.00	944,748.25	

SN	Item Description	Unit	No.	No.	L	B	H	Qty.	Rate as per CCL SOR	Amount	Remarks
4.	Dry lean cement concrete sub-base (construction of dry lean concrete sub-base over a prepared sub-grade with coarse and fine aggregate conforming to IS:383, the size of coarse aggregate not exceeding 25mm, aggregate cement ratio not to exceed 15:1 aggregate gradation after blending to be as per table 600-1, cement content not be less than 150 kg/cum, optimum moisture content to be determined during trial length construction, concrete strength not be less than 10 Mpa at 7 days, mixed in a batching plant with a paver with electronic sensor, compacting with 8-10 tonnes vibratory roller, finishing and curing.	cum			1820.00	10.50	0.15	2866.50	3,632.00	10,411,128.00	6.1 (Road works)
5.	Cement concrete Pavement (Construction of un-reinforced, dowel jointed, plain cement concrete pavement over a prepared sub base with 43 grade cement @ 400 kg per cum, coarse and fine aggregate conforming to IS 383, maximum size of coarse aggregate not exceeding 25mm, mixed in a batching and mixing plant as per approved mix design, transported to site laid with a fixed form or slip form paver, spread, compacted and finished in a continous operation including provision of contraction, expansion, construction and longitudinal joints, joints filler, seperation membrane, sealant primer, joint sealant debonding skip, dowel bars tie rod, admixture as approved curing compound finishing to lines and grades as per drawing). Construction of subgrade and earthen shoulders (construction of subgrade and earthen shoulders with approved material obtained from borrow pits with all lifts & leads, transporting to site, spreading grading to required slope and compacted to	cum			1820.00	10.00	0.30	5460.00	6,338.00	34,605,480.00	6.2 (Road work)

PART-C (Hume Pipe Culvert)

S.N	Item Description	Unit	No	No	L	B	H	Qty.	Rate as per CCL SOR	Amount	Remarks.
1.	Earthwork in excavation of foundation of structures as per drawing and technical specification, including setting out construction shoring and bracing removal of stumps and other deleterious matter, dressing of sides and bottom and backfilling with approved material.										12.1 (1) (B) (i) (Road work)
		cum		2.0	5.32	3.00	0.30	9.58			(3 nos. hume pipe of 1200 mm dia and thickness 120 mm & add 0.5 mm on both side clearance, 0.3m below the existing low land)
		cum		2.00	3.88	3.00	0.30	6.98			(2 nos hume pipe of 1200 mm dia & thickness 120 mm & dia add 0.5m on both side clearance 0.3m below the Existing low land)
		cum		2.00	2.44	3.00	0.30	4.39			(1 no hume pipe of 1200 mm dia and thickness 120 mm & add 0.5 m on both side clearance, 0.3m below the existing low land).
		cum	3	4.00	3.00	0.625	0.40	9.00			For side walls.
		cum						29.95	53.00	1,587.46	
2.	Plain cement concrete 1:3:6 nominal mix in foundation with crushed stone aggregate 40mm nominal size mechanically mixed placed in foundation and compacted by vibration including curing for 14 days.										
		cum		2.00	5.32	3.00	0.15	4.79			
		cum		2.00	3.88	3.00	0.15	3.49			
		cum		2.00	2.44	3.00	0.15	2.20			
		cum	3	2.00	3.00	0.625	0.15	2.25			
		cum		4.00			0.10	12.73	6,012.00	76,508.71	



Section of Drain

I wanted to extend my heartfelt thanks to Aman Ujjwal sir for constant support and guidance during my internship at CCL. Your mentorship has played a significant role in shaping my experience and learning, and I am truly grateful for the encouragement and trust you've shown me throughout this journey.It has been a privilege to work under your guidance, and the knowledge and values I've gained will surely stay with me as I move forward.

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